

Computational Exploration of Non-Valence Anions from Biological Quinones

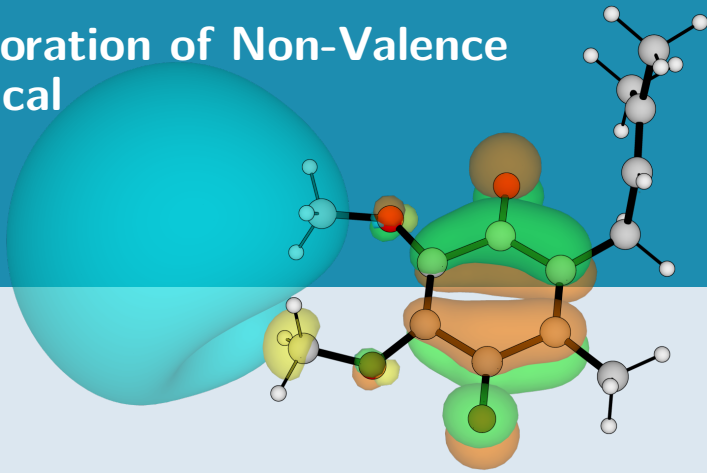
TCCM Thesis Defense

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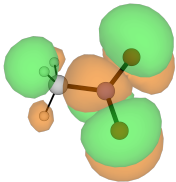
30th June 2025



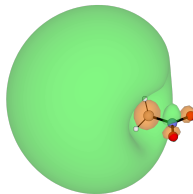
Non-valence anions

The excess electron is bound by long-range forces (e.g. dipole, quadrupole) and is located far from the molecule

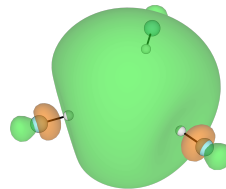
- Found in atmospheric, interstellar, and biological environments
- 'Doorway' states for electron attachment
- Large spatial extent
- Sensitive to correlation and environmental effects
- Require extremely diffuse basis sets and accurate correlation treatment



Valence-bound anion of
nitromethane



Dipole-bound anion of nitromethane

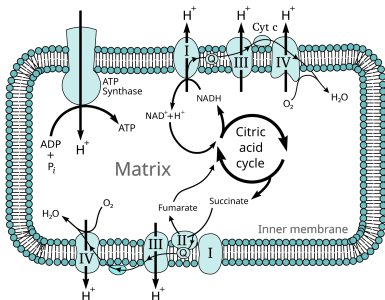


(HF)₃ solvated electron

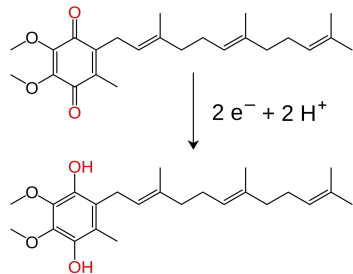
Biological Quinones: Ubiquinone (Q_n)

Quinones are essential electron carriers in biological processes

- Ubiquinone (Q_n) is a key molecule in aerobic respiration and photosynthesis
- Membrane soluble, transports electrons between enzymes
- Ubiquinone accepts two electrons and two protons to form ubiquinol



Mitochondrial Electron Transport Chain
Source: Wikimedia Commons

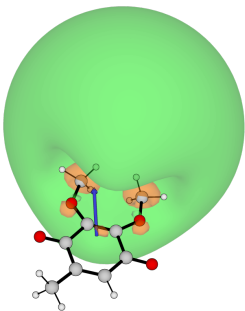


Ubiquinone reduction

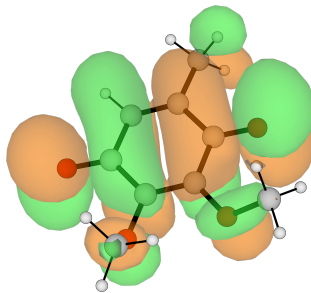
Ubiquinone Function

Ubiquinone is a lipid-soluble molecule that plays a crucial role in the electron transport chain of mitochondria.

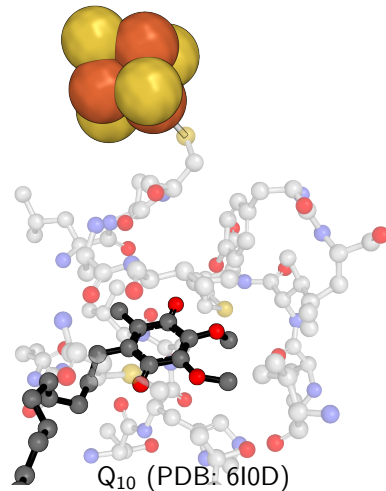
- Quinone head involved in the electron transfer
- Isoprenoid tail responsible for the solubility in the membrane
- Supports both a valence and a dipole bound anion state



Dipole-bound state



Valence-bound state



RI-CC2

Resolution-of-the-identity (RI) approximation second-order approximate coupled-cluster (CC2) method is obtained from a perturbative analysis of the CCSD model

- $|\Psi_{\text{CC}}\rangle = e^{\hat{T}}|\Psi_0\rangle$
- for CCSD and CC2: $\hat{T} = \sum_{ai} t_i^a a_a^\dagger a_i + \frac{1}{4} \sum_{aibj} t_{ij}^{ab} a_a^\dagger a_i a_b^\dagger a_j$
- CC2 double amplitudes are similar to MP2: $t_{bj}^{ai} = \frac{(ab|\hat{i}j)}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}$

Method	Single Amp.	Double Amp.	Scaling	Memory
CCSD	$\langle \Psi_i^a \bar{H} + [\bar{H}, \hat{T}_2] \Psi_0 \rangle = 0$	$\langle \Psi_{ij}^{ab} \bar{H} + [\bar{H}, \hat{T}_2]$ $+ \frac{1}{2} [[\bar{H}, \hat{T}_2], \hat{T}_2] \Psi_0 \rangle = 0$	$\mathcal{O}(N^6)$	$\mathcal{O}(N^4)$
RI-CC2	$\langle \Psi_i^a \bar{H} + [\bar{H}, \hat{T}_2] \Psi_0 \rangle = 0$	$\langle \Psi_{ij}^{ab} \bar{H} + [\hat{F}, \hat{T}_2] \Psi_0 \rangle = 0$	$\mathcal{O}(N^5)$	$\mathcal{O}(N^3)$

$$\bar{H} = e^{-\hat{T}_1} \hat{H} e^{\hat{T}_1}$$

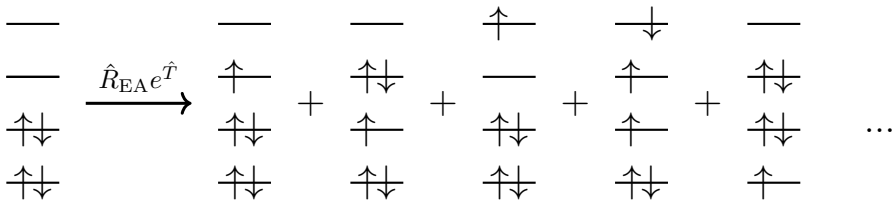
$$\hat{H} = \hat{F} + \hat{W}$$

EOM-EA

Equation-of-motion electron-attachment coupled-cluster (EOM-EA-CC) methods are well suited to study NVSs. The description is based on the wave function of the parent neutral molecule

$$|\Psi_{\text{EA}}\rangle = \hat{R}_{\text{EA}}|\Psi_{\text{CC}}\rangle$$

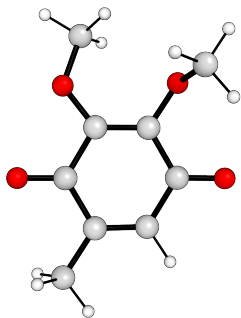
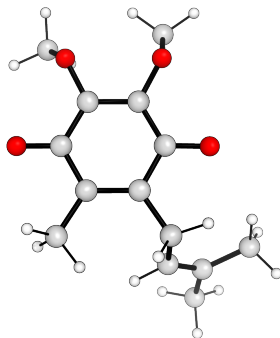
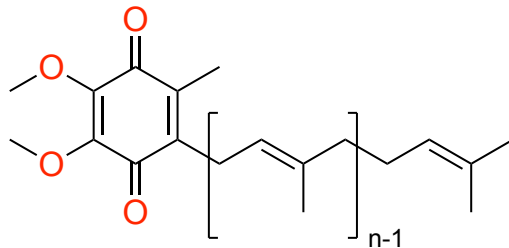
$$\hat{R}_{\text{EA}} = \sum_a r^a a_a^\dagger + \frac{1}{2} \sum_{ab} \sum_i r_i^{ab} a_a^\dagger a_i a_b^\dagger + \dots$$



Computational methods

All calculations were performed using the *Q-Chem* software.

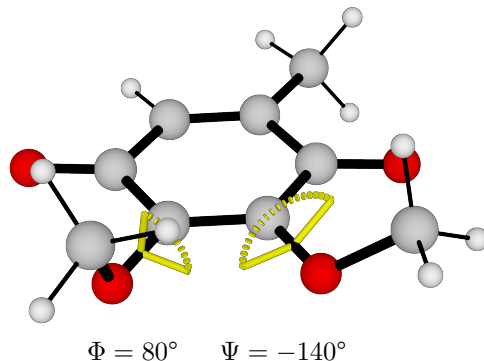
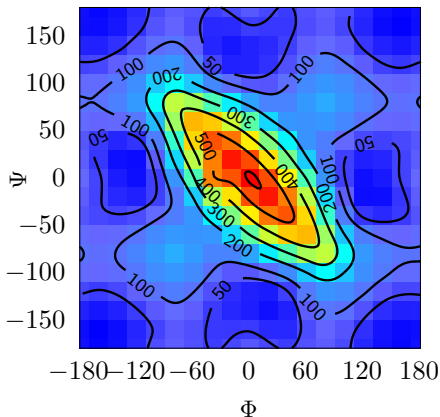
- Molecules investigated: Q_0 and Q_1
- Optimizations performed with DFT (TPSS+D3BJ/ma-def2-TZVP)
- Electron attachment energies calculated at the RI-EOM-EA-CC2/aug-cc-pVDZ+6s3p
- Implementation of Dyson orbitals at the RI-EOM-EA-CC2 level in *Q-Chem* ccmn2 module

 Q_0  Q_1  Q_n

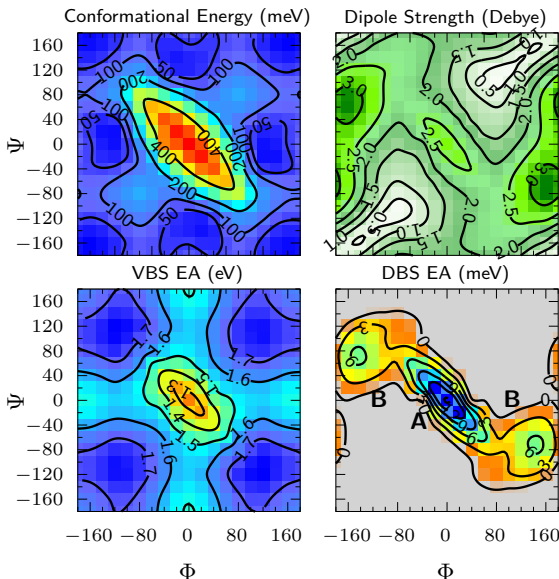
Potential Energy Surface

We can construct surfaces from the methoxy rotations of the Q₀ model.

- PES is constructed by scanning the dihedral angles Φ and Ψ
- Constrained optimisation at each point

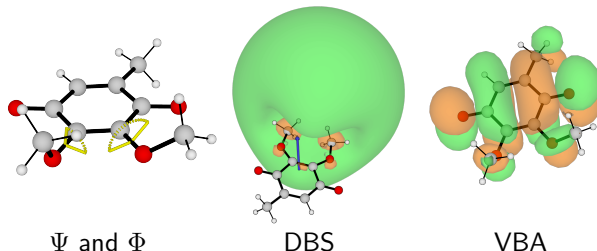


Potential Energy Surfaces of Q_0

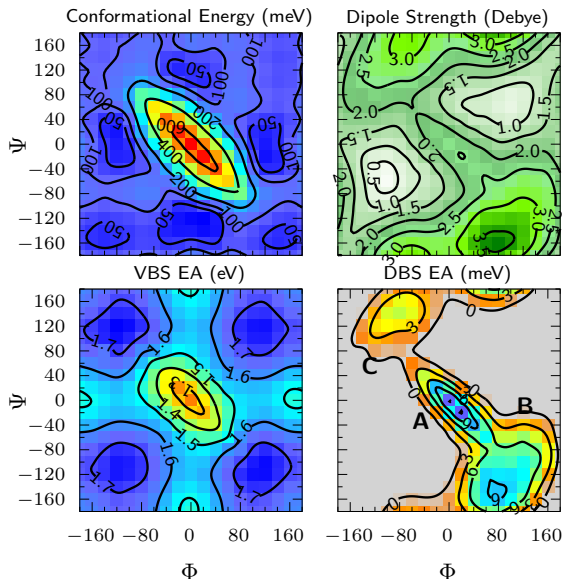


Q_0 has C_s symmetry for the methoxy rotations

- **PES:** Several minima
- **Dipole Moment:** Two strong dipole regions
- **VBS:** Dependent on electron donating or withdrawing character of the methoxy groups
- **DBS:** Follows dipole strength

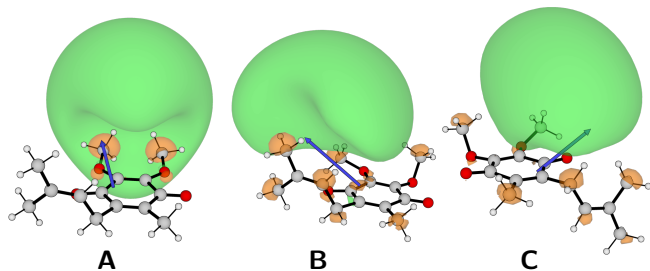


Potential Energy Surfaces of Q_1

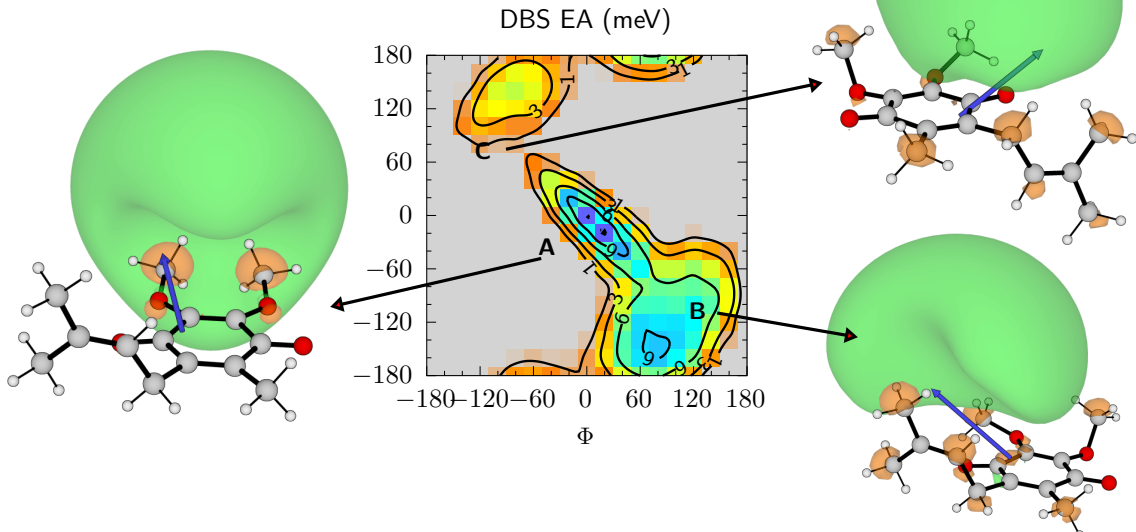


Isoprene tail breaks the symmetry

- **PES:** Isoprene tail distant, analogous to Q_0
- **Dipole Moment:** Isoprene adds fixed dipole
- **VBS:** Isoprene tail has minor effect
- **DBS:** Isoprene tail has a pronounced effect

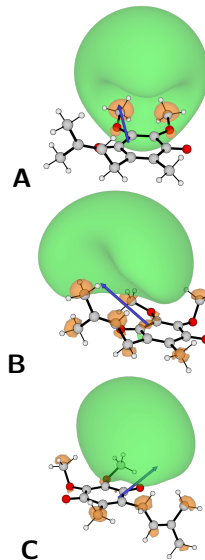
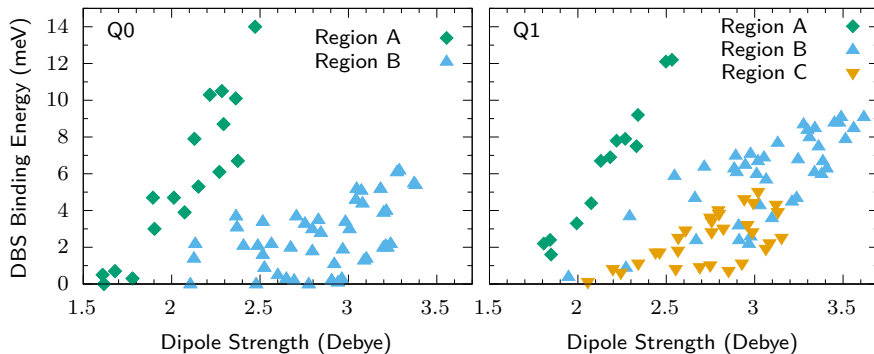


DBS Populations

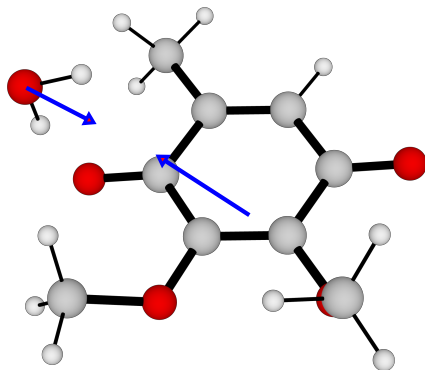


DBS Populations

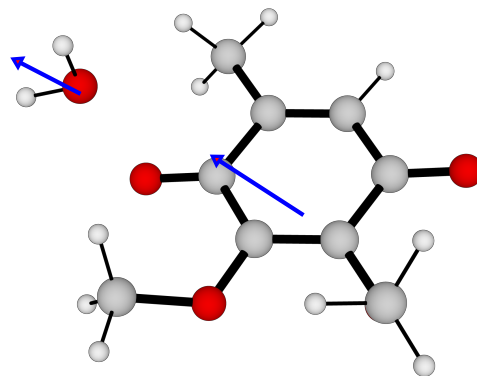
- Different regions correlate differently to the dipole strength
- **Region A**: DBS density far from other electrons
- **Regions B & C**: DBS closer to π system



Interaction with solvent molecules



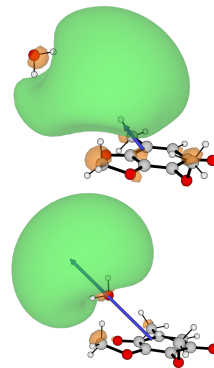
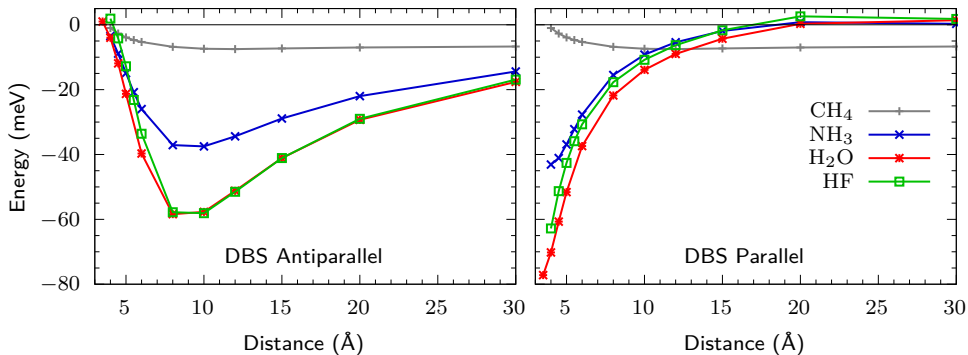
'Antiparallel'



'Parallel'

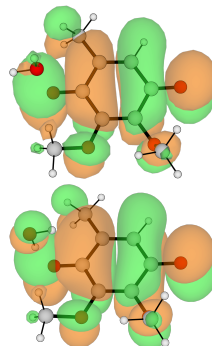
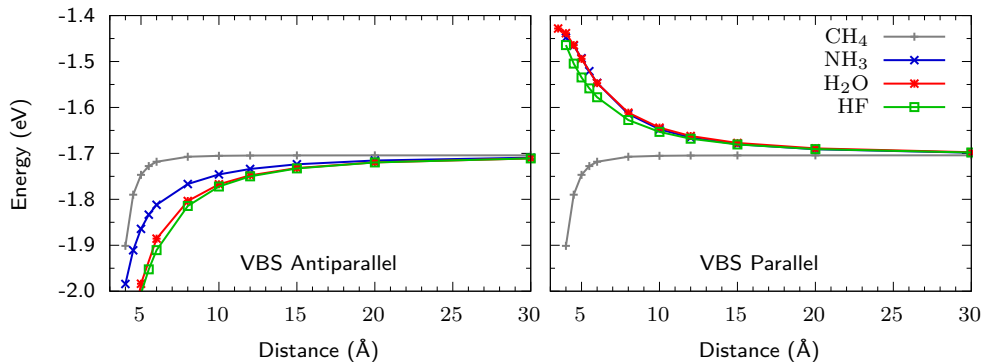
Interaction with solvent molecules (DBS)

- Binding energy increases tenfold
- **Antiparallel dipoles:** Potential well
- **Aligned dipoles:** Repulsion at large distances. At short range, dipoles add up



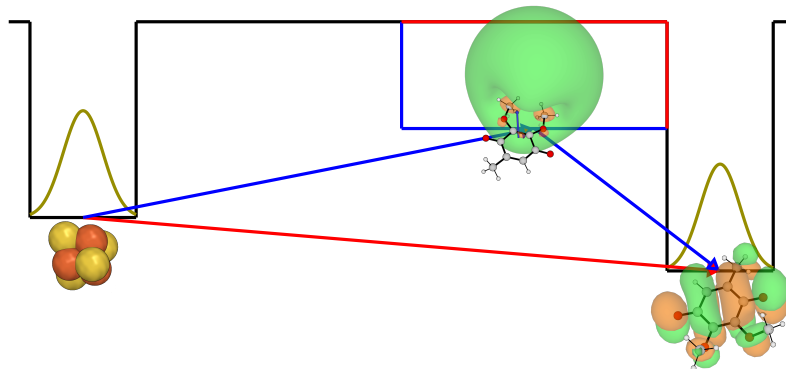
Interaction with solvent molecules (VBS)

- Intermolecular interactions larger effect than methoxy chains rotation
- **Antiparallel dipoles:** VBS strongly stabilised at short distances
- **Aligned dipoles:** VBS destabilised



Conclusions

- Methoxy chain rotation in Q_0 and Q_1 controls dipole moment and DBS energies
- In ubiquinone intermolecular interactions have a larger effect on VBS than methoxy chain rotations
- Non-valence states could play a role in long-range biological electron transfer
- Future steps: collaborative residue interactions supporting DBS



Acknowledgements



"la Caixa"



Master Studies in
Theoretical Chemistry and
Computational Modelling

VLAAMS
SUPERCOMPUTER
CENTRUM



Vlaanderen
is supercomputing



DBS Basis set convergence

Table: Electron affinity of dipole-bound radical anions computed using different augmented Dunning basis sets and EOM-EA RI-CC2 and EOM-EA RI-CCSD. Koopman' theorem (KT), and dipole moment, μ , calculated at the HF/aug-cc-pVTZ+6s3p level, and mean absolute error (MAE) taking CCSD/aug-cc-pVTZ+6s3p as reference are also given. The values are in meV and Debye respectively.

Molecule		RI-CC2						RI-CCSD		KT	μ
		2s1p	aug-cc-pVTZ		8s4p	pVDZ	pVQZ	pVDZ	pVTZ		
			4s2p	6s3p		6s3p	6s3p	6s3p	6s3p		
Acetaldehyde	CH ₃ CHO	-156.7	-27.8	-3.2	0.8	-4.6	-3.2	-4.6	-3.1	-0.4	3.29
Acetone	(CH ₃) ₂ CO	-114.9	-16.8	1.3	3.3	-0.3	0.9	-0.5	0.9	-5.1	3.46
Acetonitrile	CH ₃ CN	-61.2	12.6	19.9	20.1	18.2	20.3	17.1	18.4	4.2	4.29
Benzaldehyde	C ₆ H ₅ CHO	-97.1	-2.1	8.9	9.6	7.4	9.1	3.4	4.6	-4.9	3.77
N,N-Dimethylformamide	(CH ₃) ₂ NCHO	-81.1	5.4	14.1	14.4	13.2	14.4	13.3	13.7	1.9	4.48
DMSO	(CH ₃) ₂ SO	-84.5	4.0	15.4	16.1	14.8	15.5	14.7	14.9	2.1	4.63
Formamide	CH ₃ NO	-92.2	1.1	16.2	17.2	15.1	17.0	15.1	15.9	3.4	4.28
Methylisocyanide	CH ₃ NC	-95.1	-0.5	10.0	10.5	9.5	10.1	8.8	9.0	-1.8	3.59
Nitrobenzene	C ₆ H ₅ NO ₂	-63.6	30.6	34.8	34.8	32.5	—	25.0	25.9	5.4	5.15
Nitromethane	CH ₃ NO ₂	-82.9	5.7	14.2	14.7	13.0	14.7	12.9	13.7	3.5	4.10
Nitrosobenzene	C ₆ H ₅ NO	-125.0	1.0	11.4	—	9.9	—	5.1	6.0	-4.1	3.73
Phenylisocyanide	C ₆ H ₅ NC	-82.7	8.6	16.3	16.5	15.2	16.7	9.0	9.2	-4.9	3.61
Pyridazine	C ₄ H ₄ N ₂	-80.7	20.5	26.3	26.4	25.0	26.7	18.6	19.1	1.7	4.41
Vinylene carbonate	C ₃ H ₂ O ₃	-82.5	20.9	27.2	27.4	26.4	27.7	25.1	25.5	10	5.05
MAE		105.3	8.8	2.8	3.4	2.3	2.4	0.8	0.0	12.0	